



Haruki Watanabe

Professor of Physics & IAS Professor

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Academic Appointments

Mar. 2026 – Present: Professor of Physics, Department of Physics, and IAS Professor, HKUST Jockey Club Institute for Advanced Study, Hong Kong University of Science and Technology

Jan. 2019 – Feb. 2026: Associate Professor (PI), Department of Applied Physics, University of Tokyo

Feb. 2016 – Dec. 2018: Lecturer (PI), Department of Applied Physics, University of Tokyo

Jul. 2015 – Jan. 2016: Pappalardo Postdoctoral Fellow, Massachusetts Institute of Technology

Education

Aug. 2011 – May 2015: Ph.D. in Physics, University of California, Berkeley (Advisor: Prof. Ashvin Vishwanath)

Apr. 2010 – Mar. 2012: M.Sc. in Physics, University of Tokyo (with honors)

Apr. 2006 – Mar. 2010: B.A. in Physics, University of Tokyo (with honors)

Research Interests

Uncovering universal properties of quantum many-body systems from a symmetry perspective. My research spans several interconnected areas:

- *Spontaneous symmetry breaking and Nambu–Goldstone bosons* — Established a unified counting rule for Nambu–Goldstone bosons in nonrelativistic systems [[PRL 108, 251602 \(2012\)](#); [PRX 4, 031057 \(2014\)](#)], and proved the absence of quantum time crystals [[PRL 114, 251603 \(2015\)](#)].
- *Topological phases and symmetry indicators* — Developed a symmetry-based framework for diagnosing band topology in all 230 (and 1651 magnetic) space groups [[Nat. Commun. 8, 50 \(2017\)](#); [Sci. Adv. 4, eaat8685 \(2018\)](#)], extended to topological superconductors [[Sci. Adv. 6, eaaz8367 \(2020\)](#)].
- *Nonlinear electromagnetic responses in superconductors* — Formulated gauge-invariant theory of optical responses and established rigorous sum rules for nonlinear conductivities [[PRB 102, 165137 \(2020\)](#)].
- *Frustration-free quantum many-body systems* — Proved rigorous bounds on dynamical critical exponents and finite-size gaps in gapless frustration-free systems [[PRX 15, 041050 \(2025\)](#); [PRL 133, 176001 \(2024\)](#)].

Selected Awards

- [Particle Physics Medal](#) (Japan Particle and Nuclear Theory Forum, 2025)
- [Faculty of Engineering Best Teaching Award](#) (University of Tokyo, 2024)
- [The Commendation for Science and Technology](#) (Minister of Education, Japan, 2024)
- [New Horizons in Physics Prize](#) (Breakthrough Prize Foundation, 2022)

- Condensed-Matter Science Prize (2017)
- Yukawa Memorial Prize (2016)
- Young Scientist Award (Physical Society of Japan, 2016)
- Seitaro Nakamura Prize (2013)
- Outstanding Graduate Student Instructor (UC Berkeley, 2012)

Publications

Over 60 refereed publications (see attached publication list), including 14 in Physical Review Letters, 5 in Physical Review X, 3 in Science Advances, 2 in Proceedings of the National Academy of Sciences, and 1 in Nature Communications. Seven papers selected as Editors' Suggestions. Author of an invited review article in Annual Review of Condensed Matter Physics [**11**, 169 (2020)].

Selected Publications (asterisk denotes corresponding author; underline denotes myself)

- [1] R. Masaoka, T. Soejima, and H. Watanabe^{*}, *Rigorous lower bound of dynamic critical exponents in critical frustration-free systems*, [Phys. Rev. X **15**, 041050 \(2025\)](#).
- [2] H. Watanabe^{*}, H. Katsura, and J. Y. Lee, *Critical Spontaneous breaking of U(1) symmetry at zero temperature in one dimension*, [Phys. Rev. Lett. **133**, 176001 \(2024\)](#). [Editors' Suggestion](#)
- [3] F. Tang, S. Ono, X.-G. Wan, and H. Watanabe^{*}, *High-throughput Investigations of Topological and Nodal Superconductors*, [Phys. Rev. Lett. **129**, 027001 \(2022\)](#). [Editors' Suggestion](#)
- [4] H. Watanabe and H. C. Po^{*}, *Fractional Corner Charge of Sodium Chloride*, [Phys. Rev. X **11**, 041064 \(2021\)](#).
- [5] S. Ono, H. C. Po, and H. Watanabe^{*}, *Refined symmetry indicators for topological superconductors in all space groups*, [Sci. Adv. **6**, eaaz8367 \(2020\)](#).
- [6] H. C. Po, H. Watanabe, and A. Vishwanath^{*}, *Fragile Topology and Wannier Obstructions*, [Phys. Rev. Lett. **121**, 126402 \(2018\)](#). [Editors' Suggestion](#)
- [7] E. Khalaf, H. C. Po, A. Vishwanath, and H. Watanabe^{*}, *Symmetry indicators and anomalous surface states of topological crystalline insulators*, [Phys. Rev. X **8**, 031070 \(2018\)](#).
- [8] H. Watanabe and M. Oshikawa^{*}, *Inequivalent Berry phases for the bulk polarization*, [Phys. Rev. X **8**, 021065 \(2018\)](#).
- [9] H. C. Po, A. Vishwanath^{*}, and H. Watanabe, *Symmetry-based indicators of band topology in the 230 space groups*, [Nat. Commun. **8**, 50 \(2017\)](#).
- [10] H. Watanabe and M. Oshikawa^{*}, *Absence of Quantum Time Crystals*, [Phys. Rev. Lett. **114**, 251603 \(2015\)](#).
- [11] H. Watanabe and H. Murayama^{*}, *Effective Lagrangian for Nonrelativistic Systems*, [Phys. Rev. X **4**, 031057 \(2014\)](#).
- [12] H. Watanabe and H. Murayama^{*}, *Unified Description of Nambu–Goldstone Bosons without Lorentz Invariance*, [Phys. Rev. Lett. **108**, 251602 \(2012\)](#). [Editors' Suggestion](#)

Invited Talks

Over 40 invited talks at international workshops and conferences, including Aspen Center for Physics, ICTS Bangalore, Kavli IPMU, Kavli ITS Beijing, OIST, and UC Berkeley. Three colloquium talks, including at Caltech.

Recent Invited Talks (selected)

- International Workshop on Quantum Geometry, RIKEN (Feb. 2026)
- Generalised Symmetries and Anomalies in Quantum Phases of Matter, ICTS Bangalore (Jan. 2026)
- Mathematical Foundations of Topological Materials, IAS HKUST (Jan. 2026)
- 100 Years of Quantum Physics, ICISE Quy Nhon (Oct. 2025)
- Aspects of Generalized Symmetries, OIST (Jun. 2025)
- Unraveling the Particle World and the Cosmos, UC Berkeley (Sep. 2024)
- Recent Developments and Challenges in Topological Phases, YITP Kyoto (Jun. 2024)

Research Grants

- JSPS Grant-in-Aid for Scientific Research (B), US\$123.9k (2024 – 2028)
- JSPS Grant-in-Aid for Scientific Research (B), US\$115.3k (2021 – 2024) *Co-Investigator
- JSPS Grant-in-Aid for Scientific Research (B), US\$107.5k (2020 – 2024)
- Japan Science and Technology PRESTO, US\$200k (2018 – 2021)
- JSPS Grant-in-Aid for Young Scientists (B), US\$27.7k (2017 – 2019)
- Grant for the University of Tokyo Excellent Young Researchers, US\$40k (2017 – 2018)

(Currency conversion assumes approx. 1 USD = 150 JPY)

(Co-)Organized Workshops

- Topology, Entanglement, and Dynamics in Quantum Many-Body Systems, Dec. 2025, ISSP Kashiwa
<http://www.phys.keio.ac.jp/members/furukawa/2025topoent/>
- Recent Developments on Multipole Moments in Quantum Systems, May 2020, via Zoom
<https://sites.google.com/g.ecc.u-tokyo.ac.jp/workshop-multipole/>
- Symmetry and Topology in Condensed-Matter Physics, Jun. 2018, University of Tokyo
<https://sites.google.com/site/symmetrytopologycmp/>